

Shivam Goel

goelshivam1210@gmail.com | [personal website](#) | [linkedin](#) | [github](#) | [twitter](#) | [scholar](#)

RESEARCH INTERESTS

- Reinforcement Learning
- Contact-Rich Manipulation
- Object-Centric and Robot-Agnostic Learning
- Task and Motion Planning (TAMP)
- Sim-to-Real Transfer
- Cognitive and Neurosymbolic Robotic Architectures

EDUCATION

Tufts University

Ph.D. in Computer Science

Advisors: Prof. Jivko Sinapov & Prof. Matthias Scheutz

Medford, MA

Jan 2020 – May 2026

Washington State University

Masters of Science in Computer Science

Advisor: Prof. Matthew E. Taylor

Pullman, WA

Aug 2015 – May 2017

Uttar Pradesh Technical University

Bachelors of Engineering in Information Technology

Noida, UP

Aug 2011 – June 2015

EXPERIENCE

Mulip Lab – Tufts University

Researcher · Advisor: Prof. Jivko Sinapov

Medford, MA

Jan 2020 – May 2026

- Developed and deployed deep learning-based, contact-rich manipulation policies for doors, drawers, and large-object interaction on UR5e, Kinova arms, and Boston Dynamics Spot.
- Designed robot-agnostic learning and control pipelines enabling transfer across arms, objects, and environments.
- Integrated reinforcement learning, force/torque sensing, and motion planning into full execution stacks.
- Validated systems in simulation (MuJoCo, PyBullet) and shipped to physical robots.
- Led system design resulting in first-author publications at ICRA-25, AAMAS-24, ICDL-22.

HRI Lab – Tufts University

Researcher · Advisor: Prof. Matthias Scheutz

Medford, MA

May 2023 – May 2026

- Developed neurosymbolic robotic systems integrating learned manipulation policies with task-and-motion planning.
- Designed robot-agnostic, force-based manipulation behaviors for non-prehensile and multi-robot interaction.
- Integrated learned policies into full robotic execution stacks operating under novel objects and dynamics.
- Publications at AIJ, AAMAS; ongoing submissions to IROS-26

IRL Lab – Washington State University

Researcher · Advisor: Prof. Matthew E. Taylor

Pullman, WA

Oct 2016 – Dec 2019

- Autonomous UAV system for bird detection and deterrence in agricultural environments using vision-based perception and closed-loop control (publications at ASABE).

CASAS Lab – Washington State University

Researcher · Advisor: Prof. Diane J. Cook

Pullman, WA

Jan 2018 – Dec 2019

- Robotic assistive systems for activities of daily living, integrating object detection and ROS-based deployment of custom turtlebot based robot in smart-home environments (Journal publication in Cognitive Systems Research).

Center of AI for Social Good – University of Southern California

Research Intern · Advisor: Prof. Milind Tambe & Prof. Bistra Dilkina

Los Angeles, CA

Apr 2018 – Aug 2018

- Applied AI to wildlife conservation, developing UAV-based thermal vision methods for poacher detection.
- Increased detection accuracy from 30% to 68% improving Faster RCNN models for object detection.

ACTIVE PROJECTS

Generalizable Robotic Skills | *Physics-based Learning, Object-centric RL*

- Develop advanced motion primitives (e.g., twisting, non-prehensile object manipulation, dual-robot arms) to enable transferable robot skills across tasks and platforms.
- Paper under review at **IROS-26**

Scalable Frameworks for General-Purpose Robots | *Neurosymbolic AI, General-purpose Robotics*

- Leverage hybrid neurosymbolic architectures to integrate force-based robot-agnostic skills into larger robotic platforms capable of broad generalization.
- Journal in preparation to **Robotics Science & Systems**

Reasoning for Open-World Adaptation | *Novelty Handling, Affordances discovery*

- Incorporate symbolic reasoning to guide skill learning when encountering novel objects and their affordances.
- Preparation for submission to **CoRL-2026**

SYSTEMS & DEPLOYMENT EXPERIENCE

- Deployed learning-based manipulation systems on UR5e, Kinova arms, and Boston Dynamics Spot.
- Built robotic stacks using ROS/ROS2, across mobile and manipulator platforms.
- Experienced debugging hardware–software integration failures in real-world environments.

PUBLICATIONS

Open-World Novelty Handling and Adaptation

- **Shivam Goel**, Panagiotis Lymperopoulos, Ravenna Thielstrom, Evan Krause, Patrick Feeney, Pierrick Lorang, Sarah Schneider, Yichen Wei, Eric Kildebeck, Stephen Goss, Michael C. Hughes, Liping Liu, Jivko Sinapov, Matthias Scheutz. [A Neurosymbolic Cognitive Architecture Framework for Handling Novelties in Open Worlds](#). **Artificial Intelligence Journal (AIJ)**, 2024.
- **Shivam Goel**, Yash Shukla, Vasanth Sarathy, Matthias Scheutz, Jivko Sinapov. [RAPid-Learn: A Framework for Learning to Recover for Handling Novelties in Open-World Environments](#). **ICDL**, 2022.

Sim-to-Real and Contact-Rich Manipulation

- **Shivam Goel**, Shijie Fang, Wenchang Gao, Matthias Scheutz, Jivko Sinapov. [FLEX: A Framework for Learning Robot-Agnostic Force-based Skills Involving Sustained Contact Object Manipulation](#). **ICRA**, 2025.
- Yash Shukla, Bharat Kesari, **Shivam Goel**, Robert Wright, Jivko Sinapov. [Few-Shot Policy Transfer through Observation Mapping and Behavior Cloning](#). **IROS**, 2023.

Task and Motion Planning and World Models

- Pierrick Lorang, **Shivam Goel**, Yash Shukla, Patrick Zips, Matthias Scheutz. [A Framework for Neurosymbolic Goal-Conditioned Continual Learning for Open World Environments](#). **IROS**, 2024.
- James Staley, **Shivam Goel**, Yash Shukla, Elaine Schaertl Short. [Agent-Centric Human Demonstrations Train World Models](#). **Reinforcement Learning Conference (RLC)**, 2024.

Earlier Work: Computer Vision and Healthcare/Assistive Robotics

- Garrett Wilson, Christopher Pereyda, Nisha Raghunath, Gabriel de la Cruz, **Shivam Goel**, Sepehr Nesaei, Bryan Minor, Maureen Schmitter-Edgecombe, Matthew E. Taylor, Diane J. Cook. [Robot-Enabled Support of Daily Activities in Smart Home Environments](#). **Cognitive Systems Research**, 2019.
- Santosh Bhusal, Kapil Khanal, **Shivam Goel**, Manoj Karkee, Matthew E. Taylor. [Bird Deterrence in a Vineyard Using an Unmanned Aerial System](#). *Transactions of the ASABE*, 2019.
- **Shivam Goel**, Santosh Bhusal, Matthew E. Taylor, Manoj Karkee. [Detection and Localization of Birds for Bird Deterrence Using UAS](#). **ASABE Annual Meeting**, 2017.
- Santosh Bhusal, **Shivam Goel**, Kapil Khanal, Matthew E. Taylor, Manoj Karkee. [Bird Detection, Tracking and Counting in Wine Grapes](#). **ASABE Annual Meeting**, 2017.

TECHNICAL DEMONSTRATIONS

Tufts Robotics Open House | 2024, 2025

- Showcased neurosymbolic robotic learning systems on quadruped legged robots (Spot) and wheeled mobile manipulators (LoCoBot) for novelty handling to cross-disciplinary audience.
- Demonstrated Robotic System’s ability to take voice commands, reason and act in the world.

MassRobotics | Boston, 2023

- Demonstrated VR-based interaction capabilities of a turtlebot

Tufts Graduate Showcase | 2022

- Presented multimodal robotic interaction systems to cross-disciplinary audiences.

TECHNICAL EXPERTISE

Robotics: Contact-rich manipulation, whole-body control primitives, force/torque learning, sim-to-real transfer, task-and-motion planning

Robots: UR5e, Kinova, Boston Dynamics Spot, LoCoBot (5-DOF), TurtleBot

Simulation: MuJoCo, PyBullet, Gazebo, Robosuite

Learning: PyTorch; RL/IL pipelines; OpenAI Gym-style environments

Middleware: ROS/ROS2

Programming: Python ,C/C++, Java, MATLAB (intermediate)

Last updated: May 2026